통계열역학 (2009년도 1학기)

2차 과제 정답 (**총 65 점**)

1. **(10점)**

x	1	2	3	4	5
$p(x)$	0.10	0.15	0.05	0.50	0.20

$$\begin{aligned}
\langle x\rangle &= \sum_{j=1}^{n} x_j p(x_j) \\
&= (1)(0.10)+(2)(0.15)+(3)(0.05)+(4)(0.50)+(5)(0.20) = 3.55 \\
\langle x^2\rangle &= \sum_{j=1}^{n} x_j^2 p(x_j) \\
&= (1)(0.10)+(4)(0.15)+(9)(0.05)+(16)(0.50)+(25)(0.20) = 14.14 \\
\sigma^2 &= \langle x^2\rangle - \langle x\rangle^2 \\
&= 14.14 - 3.55^2 = 1.54 \\
\sigma &= \sqrt{1.54} = 1.24
\end{aligned}$$

2. **(총 10 점, 각 5 점)**

(a) 정규화 조건에 의해 $\int_{-\infty}^{\infty} p(x)dx = 1$ 이므로

$$\int_{-\infty}^{\infty} p(x)dx = \int_a^b A dx = 1 \Rightarrow A\Big[x\Big]_a^b = 1 \Rightarrow A = \frac{1}{b-a}$$

(b)

$$\begin{aligned}
\langle x\rangle &= \int_{-\infty}^{\infty} x p(x)dx \\
&= \frac{1}{b-a}\int_a^b x dx = \frac{1}{b-a}\left[\frac{1}{2}x^2\right]_a^b = \frac{1}{2(b-a)}\left(b^2-a^2\right) = \frac{1}{2}(a+b)
\end{aligned}$$

$$\begin{aligned}
\langle x^2\rangle &= \int_{-\infty}^{\infty} x^2 p(x)dx \\
&= \frac{1}{b-a}\int_a^b x^2 dx = \frac{1}{b-a}\left[\frac{1}{3}x^3\right]_a^b = \frac{1}{3(b-a)}\left(b^3-a^3\right) = \frac{1}{3}(a^2+ab+b^2)
\end{aligned}$$

$$\begin{aligned}\sigma_x^2 &= \langle x^2\rangle - \langle x\rangle^2 \\ &= \frac{1}{3}\big(a^2+ab+b^2\big) - \left(\frac{1}{2}\big(a+b\big)\right)^2 \\ &= \frac{1}{12}\big(4a^2+4ab+4b^2-3a^2-6ab-3b^2\big) = \frac{1}{12}\big(a^2-2ab+b^2\big) = \frac{1}{12}\big(b-a\big)^2\end{aligned}$$

3. **(총 30점, 각 5점)**

(a) 분자분배함수는 다음과 같이 쓸 수 있으므로

$$q(V,T) = \sum_E \Omega(E) e^{-\beta E} = 1 + e^{-\beta\epsilon}$$

분자들이 서로 구별가능한 경우에 정준분배함수는

$$Q(N,V,T) = q^N = \big(1+e^{-\beta\epsilon}\big)^N$$

(b) Helmholtz 자유에너지는

$$A(N,V,T) = -k_BT\ln Q(N,V,T) = -k_BT\ln\big(1+1e^{-\beta\epsilon}\big)^N = -Nk_BT\ln\big(1+e^{-\beta\epsilon}\big)$$

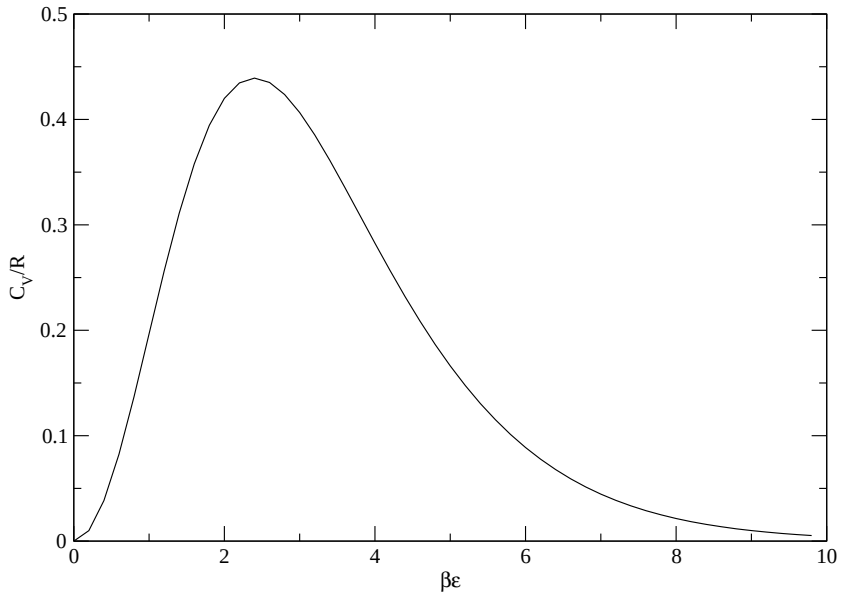
(c) 평균에너지 $\langle E\rangle$ 는

$$\begin{aligned}\langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,V} \\ &= k_BT^2\frac{\partial}{\partial T}\ln\Big[\big(1+e^{-\beta\epsilon}\big)^N\Big] \\ &= Nk_BT^2\frac{\partial}{\partial T}\ln\big(1+e^{-\epsilon/k_BT}\big) \\ &= Nk_BT^2\frac{\frac{\epsilon}{k_BT^2}e^{-\epsilon/k_BT}}{1+e^{-\epsilon/k_BT}} \\ &= \frac{N\epsilon e^{-\epsilon/k_BT}}{1+e^{-\epsilon/k_BT}}\end{aligned}$$

(d) 일정부피 몰열용량 $\bar{C}_V$ 는

$$\begin{aligned}
\bar{C}_V &= \left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N_A,V} \\
&= \frac{\partial}{\partial T}\left[\frac{N_A\epsilon e^{-\epsilon/k_BT}}{1+e^{-\epsilon/k_BT}}\right] \\
&= N_A\epsilon\left[\frac{\frac{\epsilon}{k_BT^2}e^{-\epsilon/k_BT}(1+e^{-\epsilon/k_BT}) - e^{-\epsilon/k_BT}\frac{\epsilon}{k_BT^2}e^{-\epsilon/k_BT}}{(1+e^{-\epsilon/k_BT})^2}\right] \\
&= \frac{N_A\epsilon^2}{k_BT^2}\frac{e^{-\epsilon/k_BT}}{(1+e^{-\epsilon/k_BT})^2} \\
&= N_Ak_B\frac{\epsilon^2}{k_B^2T^2}\frac{e^{-\epsilon/k_BT}}{(1+e^{-\epsilon/k_BT})^2} \\
&= R\left(\frac{\epsilon}{k_BT}\right)^2\frac{e^{-\epsilon/k_BT}}{(1+e^{-\epsilon/k_BT})^2} \\
&= R\big(\beta\epsilon\big)^2\frac{e^{-\beta\epsilon}}{(1+e^{-\beta\epsilon})^2} \qquad (1)
\end{aligned}$$

(e) (1) 식으로부터



4. **(총 15점, 각 5점)**

(a) Helmholtz 자유에너지는

$$\begin{aligned} A(N,V,T) &= -k_BT\ln Q(N,V,T) \\ &= -k_BT\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/(Vk_BT)}\right] \\ &= -k_BT\left[\frac{3N}{2}\ln\left(\frac{2\pi mk_BT}{h^2}\right)+N\ln(V-Nb)+\frac{aN^2}{Vk_BT}-\ln N!\right] \end{aligned}$$

(b) 평균에너지 $\langle E\rangle$ 는

$$\begin{aligned} \langle E\rangle &= k_BT^2\left(\frac{\partial\ln Q}{\partial T}\right)_{N,V} \\ &= k_BT^2\frac{\partial}{\partial T}\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/(Vk_BT)}\right] \\ &= k_BT^2\frac{\partial}{\partial T}\left\{-\ln N!+\frac{3N}{2}\ln\left(\frac{2\pi mk_B}{h^2}\right)+\frac{3N}{2}\ln T+N\ln(V-Nb)+\frac{aN^2}{Vk_BT}\right\} \\ &= \frac{3}{2}Nk_BT-\frac{aN^2}{V} \end{aligned}$$

C_V 는

$$C_V=\left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N,V}=\frac{3}{2}Nk_B$$

(c)

$$\begin{aligned} P &= -\left(\frac{\partial A}{\partial V}\right)_{N,T} \\ &= k_BT\left(\frac{\partial\ln Q}{\partial V}\right)_{N,T} \\ &= k_BT\frac{\partial}{\partial V}\left\{\ln\left[\frac{1}{N!}\left(\frac{2\pi mk_BT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/(Vk_BT)}\right]\right\} \\ &= k_BT\frac{\partial}{\partial V}\left\{-\ln N!+\frac{3N}{2}\ln\left(\frac{2\pi mk_BT}{h^2}\right)+N\ln(V-Nb)+\frac{aN^2}{Vk_BT}\right\} \\ &= k_BT\left\{\frac{N}{V-Nb}-\frac{aN^2}{V^2k_BT}\right\} \\ &= \frac{Nk_BT}{V-Nb}-\frac{aN^2}{V^2} \end{aligned}$$

5. **(5점)** 세 가지 미시상태 1,2,3중 하나로 존재할 수 있는 열역학 계가 각 상태로 존재할 확률은

$$p_j = \frac{e^{-\epsilon_j/k_BT}}{q} \tag{2}$$

위 식에서 $q = e^{-\epsilon_1/k_TT} + e^{-\epsilon_2/k_BT} + e^{-\epsilon_3/k_BT}$. 상태 1의 에너지는 0이므로 $p_1 = 1/q$이고

$$\begin{aligned} p_1 &= \frac{1}{q} = 0.9 \\ &\Rightarrow \quad q = \frac{1}{0.9} = \frac{10}{9} = 1.11. \end{aligned}$$

(2) 식을 이용하면,

$$\epsilon_j = -k_BT\ln(p_jq).$$

그러므로

$$\begin{aligned} \epsilon_2 &= -k_BT\ln(p_2q) = -k_BT\ln(0.09\times 0.11) = 0.954\times 10^{-20}\ \text{J}, \\ \epsilon_3 &= -k_BT\ln(p_3q) = -k_BT\ln(0.01\times 0.11) = 1.86\times 10^{-20}\ \text{J}. \end{aligned}$$